

RIDA MELKAOUI

INDUSTRIAL ENGINEER | DATA, AI & BUSINESS INTELLIGENCE

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LinkedIn | GitHub | Portfolio



PROFILE

Industrial & Production Engineer who combines supplier-quality ownership with data analytics, BI and applied AI. At Magna, leads electronics and metal supplier quality across international programs and builds the operating views used to track launch readiness, issues and actions. Translates industrial problems into measurable decision products with Python, SQL and Power BI.

PROFESSIONAL EXPERIENCE

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| <b>MAGNA INTERNATIONAL</b><br><i>Supplier Quality Engineer — Electronics &amp; Metal Commodities</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Kenitra, Morocco<br>Mar 2026 — Present     |
| <ul style="list-style-type: none"><li>Own supplier-quality performance for electronics and metal commodities across China, South Korea, Spain, Slovakia, Germany and Morocco, using launch readiness, 8D/CAPA status and open-issue risk to focus escalation.</li><li>Developed X-Ray, a full SQE BI and workflow system covering nomination, Safe Launch and SOP; unified ownership, milestones, PPAP/DVP evidence, critical parts and open issues across 6 control modules and 4 lifecycle gates.</li><li>Implemented KPI views and an automated notification pipeline for missing baselines, overdue actions and gate risks, keeping responsibilities and deadlines visible across the supplier portfolio.</li></ul> |                                            |
| <b>STELLANTIS R&amp;D</b><br><i>Engineering Intern — Data Analytics, Process Optimization &amp; AI</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Casablanca, Morocco<br>Feb 2025 — Aug 2025 |
| <ul style="list-style-type: none"><li>Mapped the E-BOM/M-BOM anomaly process and built a Python application plus structured Databricks model for detection, prioritization and traceability; reduced the validated treatment cycle from 2 days to under 3 minutes (−98% manual handling, 100% project test-case reliability).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                 |                                            |
| <b>SANY — FORGE DE BAZAS</b><br><i>Engineering Intern — Process Digitalization &amp; Maintenance</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Casablanca, Morocco<br>Jul 2024 — Sep 2024 |
| <ul style="list-style-type: none"><li>Built a Unity/NLP assistant that structured access to maintenance and spare-parts knowledge, replacing fragmented shop-floor search with a guided information flow.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                            |
| <b>3D SMART FACTORY</b><br><i>Data Analyst Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mohammedia, Morocco<br>Aug 2023 — Sep 2023 |
| <ul style="list-style-type: none"><li>Structured Power BI and Excel reporting around OEE, FPY, scrap and downtime, giving Lean reviews a single view of production loss drivers and bottlenecks.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                            |

PROJECTS

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| <b>Demand / Order — Forecasting &amp; replenishment</b>                                                                                                                                                                                                                                                                                                                                      |
| Built a cutoff-safe Python forecasting and replenishment engine to decide what to stock and where to review exceptions. It modeled 900 M5 item-store series, benchmarked candidates against calibrated seasonal baselines and converted the best signal into inventory scenarios—improving WAPE by 4.5% and simulating 1,528 fewer lost units at 96.7% fulfilled demand on a 28-day holdout. |
| <b>Route / Control — Optimization &amp; operations</b>                                                                                                                                                                                                                                                                                                                                       |
| Built a dispatch control tower that learned zone-order preferences from 6,112 public histories, combined directed travel times with constrained local search and surfaced exceptions. On an unseen 13-route cohort, it cut simulated drive time by 2.8%, preserved 100% timed-package adherence and eliminated 332 zone re-entries.                                                          |
| <b>Agent / Proof — AI evaluation &amp; observability</b>                                                                                                                                                                                                                                                                                                                                     |
| Built an AI-agent evaluation lab over 3,336 official tool-use trajectories, measuring pass@k, repeatability, cost, latency and failure modes before release. Its pinned GPT-4.1 telecom cohort revealed a 14.9-point gap between one-shot and repeatable success.                                                                                                                            |

EDUCATION

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| Mohammed VI International Civil Aviation Academy (AIAC)<br><i>Engineering Degree   Industrial &amp; Production Engineering</i> | Nouaceur, Morocco<br>2022 — 2025 |
| Salmane El Farissi Preparatory Classes<br><i>Preparatory Studies   MPSI - MP</i>                                               | Salé, Morocco<br>2020 — 2022     |

SKILLS

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| <b>. DATA &amp; ANALYTICS</b>     | Python · SQL · Pandas · NumPy · scikit-learn · statistics · forecasting · experimentation · data modeling · Databricks  |
| <b>. BI &amp; DELIVERY</b>        | Power BI · DAX · Excel · Power Query · Tableau · React · TypeScript · FastAPI · Git · Docker · KPI systems · automation |
| <b>. INDUSTRIAL &amp; QUALITY</b> | Supplier/process performance · PPM · OEE · FPY · 8D/CAPA · FMEA · root-cause analysis · Lean · launch/SOP governance    |

LANGUAGES

|                  |                  |                                           |
|------------------|------------------|-------------------------------------------|
| Arabic<br>Native | French<br>Fluent | English<br>C1   TOEIC 855   EF SET 62/100 |
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CERTIFICATIONS

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|--------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------|
| IBM Data Analyst Professional Certificate<br>IBM / Coursera   2026 | EF SET English Certificate   C1<br>EF SET   62/100 | TOEIC Listening & Reading   C1<br>Score   855 |
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